

SAYON SAHA

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Project portfolio

EDUCATION

Iowa State University | Ames, IA

Expected May 2028

Dual Bachelor of Science degrees in Computer Engineering and Electrical Engineering

GPA: 3.8 | Dean's List | IEEE-HKN (Eta Kappa Nu)

TECHNICAL SKILLS

Languages: C, C++, Java, Verilog, SQL

Tools: Cadence Virtuoso, KiCad, LTspice, Quartus Prime, Questa/ModelSim, Git

Currently learning: Spring Boot, VHDL, RISC-V architecture, IC design and fabrication

ENGINEERING PROJECTS

CyBot Naval Deminer | Embedded C, TI TM4C123

Spring 2026

- Developed embedded C navigation and control software in a four-person team for a CyBot robot simulating naval mine detection, with autonomous and manual operating modes.
- Integrated ADC-based IR sensing, ultrasonic ranging, PWM-controlled 180-degree servo scans, and cliff/bump detection for object classification and obstacle avoidance.
- Implemented repeated-scan confirmation of candidate targets and interrupt-driven UART control with manual takeover and stop handling.

Desktop Speakers, Subwoofer & Audio Circuit Builds

- Designed and built two-way desktop speakers and a subwoofer; performed speaker impedance matching and tuned crossover frequencies with faculty guidance.
- Prototyped a three-way crossover network on a breadboard to separate low-, mid-, and high-frequency signal paths.
- Built TPA3122D2 class-D, class-AB, and headphone amplifiers, along with a 2.1 amplifier system for desktop audio.

CMOS IC Design & Simulation | Cadence Virtuoso

Fall 2026 - Ongoing

- Created transistor-level CMOS inverter schematics and reusable symbols; simulated DC transfer and transient responses while varying output capacitance from 1 pF to 1 uF.
- Built cascaded inverter circuits and an introductory PMOS layout; analyzed an RC low-pass filter using DC, AC, and transient simulation.

LabMate Equipment API | Java, Spring Boot

Fall 2026 - Ongoing

- Developing a team laboratory-management backend; implemented six REST endpoints for equipment CRUD, listing, and status filtering using JSON, path variables, and query parameters.

Other builds: Arduino LED maze game, ICSP shield, and time-temperature device.

EXPERIENCE AND LEADERSHIP

Academic Tutor | Iowa State University

Jan 2025 - Present

- Supported 20+ students in MATH 1430, MATH 1600, and MATH 1650 during Fall 2026; tailor explanations and practice to individual learning needs.
- Tutoring experience includes the ISU Academic Success Center and Knack, supporting chemistry, calculus, and introductory programming through individual and group sessions.

Technical Mentor | Audio & Arduino Club, IEEE at ISU

Aug 2025 - Present

- Lead electronics build sessions, support instruction with Dr. Tuttle, and prepare materials for hands-on circuit and Arduino projects.